Ying Siu (Sheryl) Liang

Ph.D.

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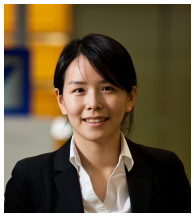

Summary

I am interested in developing generalisable solutions for autonomous robots to operate in real end-user environments. I have implemented solutions for end-user robot programming as well as for robot architecture and their belief management to control its behaviour. My ultimate goal is to create an autonomous robot that can learn new behaviours, adapt to its environment and become human's perfect assistant.

Education

2016–2019 **Ph.D. in Computer Science**, *Université Grenoble Alpes*, France.
Major in Artificial Intelligence and Robotics. Defended in June 2019
Dissertation: *End-User Robot Programming in Cobotic Environments*

2015–2016 **Master of Science (M.S.) in Informatics**, *Grenoble INP - ENSIMAG*, France.
Specialisation: Artificial Intelligence and Web
Courses: Machine Learning, Knowledge Representation & Reasoning, Multi-Agent Systems

2008–2012 **Integrated Master of Science (MSci) in Mathematics and Computer Science**, *Imperial College London*, United Kingdom, *First Class Honours*.

2000–2008 **Bundesrealgymnasium 18, Wien**, *Vienna*, Austria, Secondary school, emphasis on Geometric Modelling, IT & Communication Technology, *Graduated with straight As*.
Matura (A-Level equivalent) in Mathematics, English, Geometric Modelling, German, French

Skills

Development	C++, Python, Clojure, JavaScript, HTML, LaTeX, Git
Previous	Java, Prolog, Haskell, MATLAB, C#, Perl, PHP, SQL
OS	Ubuntu Linux, Windows, MacOS
	ROS, Visual Studio Code, OpenCV, Polymer
Robots	Universal Robot, Baxter Robot, Fetch Mobile Manipulator, PR2
Spoken languages	German (native), English (bilingual), French (C2), Chinese (C1), Spanish (A2)

Attended Courses

Q3 2021 IS5452 Affective Computing, National University of Singapore, Singapore

Aug 2018 ROS Summer School, FH Aachen, Germany

Work Experience

Jan 2020–Present **Research Scientist**, *Institute for High Performance Computing*, A*STAR, Singapore.
CollabAI project in Social and Cognitive Computing (Clojure, Python, ROS)

- Worked on the robot's cognitive architecture (ICARUS) to integrate functions from perception, manipulation, and NLU to perform high-level reasoning and control the robot's behaviour to achieve goals with Universal Robots (UR5, UR16e).
- Collaborated with other development teams to implement, test, and troubleshoot new robotic behaviours to improve the robot's capabilities [1].
- Improved the robot's belief management with a new object permanence model [2,3] that tracks invisible displacements and outperformed state-of-the-art models.

Jun–Sep 2019 **Research Intern**, *Institute for Infocomm Research*, A*STAR, Singapore.
Inferring the Geometric Nullspace for Robot Skills from Human Demonstrations [5]
- Worked on a setup to infer geometric constraints and their nullspace from human demonstrations (using OptiTrack) that can be used by the iCub robot to generalise skills to perform more complex tasks (C++).

Oct 2016 – Jun 2019 **Ph.D. candidate**, *Grenoble Informatics Lab*, Univ. Grenoble Alpes, France.
End-User Robot Programming of Action Models for Symbolic Task Planning [6,7,9]
- Designed and implemented iRoPro, an interactive Robot Programming framework for end-users to teach a robot new actions that can be reused to solve unseen problems [4].
- Integrated functions from object perception (Point Cloud segementation), manipulation (Programming by Demonstration), motion planning (MoveIt), behaviour planning (PDDL) and human–computer interaction (JavaScript/Polymer) into a full-stack software deployed on a Baxter robot (C++, ROS).
- Conducted human-robot experiments for user acceptance testing and system evaluation.

Jul–Dec 2017 **Student Intern**, *Human-Centered Robotics Lab*, Univ. of Washington, USA.
Simultaneous End-User Programming of Goals and Actions for Robotic Shelf Organisation [8]
- Learned about the specifics of the robot programming infrastructure Rapid PbD and contributed to new software releases.
- Designed and implemented a software on a Fetch mobile manipulator to learn goals and actions from human demonstrations for robotic shelf organising tasks (C++, ROS).

Feb–Jun 2016 **Student Intern**, *Grenoble Informatics Lab*, Univ. Grenoble Alpes, France.
Robot Programming by Demonstration in Cobotic Environments [10]
- Designed and implemented a robot programming by demonstration software using OpenCV and the Baxter Robot SDK (Python, ROS).
- Conducted qualitative user experiments to evaluate the software's usability with non-robotics end-users.

Sep 2012 – Sep 2015 **Software Engineer**, *Global Technology, Deutsche Bank*, London, United Kingdom.
- Developed functionalities for data retrieval from external data sources to allow users to query and modify data directly from an OLAP Cube via a web app (C#, HTML/JavaScript).

Additional Responsibilities

Jan 2021– **Team lead**, *Project on Engagement Detection in Virtual Classes*, Singapore.

Feb 2020– **Group leader**, *Managing Cognitive Systems Group*, Singapore.

April 2021 **Co-coordinator**, *NIE-ASTAR AI in Education workshop*, Singapore.

2017– **Reviewer**, *IROS'21, ICRA'18,19,21, ACII'21, RSS'19, RO-MAN'19, HRI'18*.

May–Jul 2018 **Supervision of 2 undergraduate interns**, *UI design for iRoPro*, Grenoble, France.

Feb 2017 **Teaching Assistant**, *Introduction to Databases*, Grenoble, France.

2021– **Math Tutor Volunteer**, *South Central Community Family Service Centre*, Singapore.

2020– **Trishaw Pilot Volunteer**, *Cycling Without Age*, Singapore.

2013–2015 **Math Tutor Volunteer**, *Volunteering Matters*, London, United Kingdom.

Awards and Honours

2016–2019 French Ministry PhD research grant by MSTII Doctoral School (ED MSTII)

Mar 2018 Prize for best presentation at the LIG PhD day

Jul 2017 IDEX Intl. Mobility Grant to collaborate with the Univ. of Washington

Jun 2017 Prize for best presentation at the ED MSTII PhD day

Hobbies

Basketball, Table tennis, Snowboarding, Boardgames, Cooking, Languages

Publications and Other Communication

[1] D. Choi, W. Shi, Y.S. Liang, K.H. Yeo, J.J. Kim (2021). **Controlling Industrial Robots with High-level Verbal Commands.** Intl. Conf. on Social Robotics (ICSR). Singapore: IEEE/RSJ Press. (*Accepted*)

[2] Y.S. Liang, C. Zhang, D. Choi, K. Kwok (2021). **Improving Object Permanence using Action Annotations and Reasoning.** Intl. Conf. on Intelligent Robots and Systems (IROS). Virtual (Oral): IEEE/RSJ Press.

[3] Y.S. Liang, D. Choi, K. Kwok (2021). **Maintaining a Reliable World Model using Action-aware Perceptual Anchoring.** Intl. Conf. on Robotics and Automation (ICRA). Xi'An, China (Oral): IEEE Press.

[4] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2021). **iRoPro, an interactive Robot Programming Framework.** Intl. Journal of Social Robotics (IJSR). Springer-Verlag.

[5] C. Cai, Y.S. Liang, N. Somani, Y. Wu (2020). **Inferring the Geometric Nullspace of Robot Skills from Human Demonstrations.** Intl. Conf. on Robotics and Automation (ICRA). Virtual: IEEE Press.

[6] Y.S. Liang (2019). **End-User Robot Programming in Cobotic Enviroments.** PhD thesis. Grenoble, France.

[7] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2019). **End-User Programming of Low- and High-Level Actions for Robotic Task Planning.** Intl. Symposium on Robot and Human Interactive Communication (RO-MAN). New Delhi, India (Oral): IEEE Press.

[8] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty, M. Cakmak (2018). **Simultaneous End-User Programming of Goals and Actions for Robotic Shelf Organization.** Intl. Conf. on Intelligent Robots and Systems (IROS). Madrid, Spain (Oral): IEEE/RSJ Press.

[9] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2017). **Evaluation of a Robot Programming Framework for Non-Experts using Symbolic Planning Representations.** Intl. Symposium on Robot and Human Interactive Communication (RO-MAN). Lisbon, Portugal (Poster): IEEE Press.

[10] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2017). **A Framework for Robot Programming in Cobotic Environments: First User Experiments.** Intl. Conf. on Mechatronics and Robotics Engineering (ICMRE). Paris, France (Oral): ACM.

May 2018 **Spotlight talk**, *A Robot Programming Framework for Non-Experts*, Robots in Human Environments (RHUM) Workshop, Grenoble, France.

Jun 2017 **Oral presentation**, *Robot Programming by Demonstration*, Affects, Artificial Companions and Interactions (ACAI) Working Group, Paris, France.